Functions, packages, the tidyverse & objects

Day 3 - Wednesday, May 20, 2026

Today: Wednesday May 20

Before class

- **Perusall Assignment**: *Data Computing* §3.1–3.5 (Click here).
- **HW 0: Setup confirmation** (Click here).

Plan for today

- Recap & reading R code
- Functions
- Packages and the tidyverse
- Histograms in ggplot2
- In-class activity

Helpful links

- Syllabus · AI policy · Schedule · *Data Computing* §3

Quick recap from yesterday

Yesterday we covered:

- The four panes of RStudio (Source, Console, Environment, Output)
- R as a calculator
- Data types (numeric, integer, character, logical)
- Data structures (vectors, lists, matrices, data frames, arrays)
- Creating objects with `<-`

Today we'll build on that to start *reading* and *writing* real R code.

Reading R code

Yesterday's textbook reading made one key point:

> Before you can write R code, you need to be able to **read** it.

Most people who write code start by **copying and modifying** existing code.

To do that well, you need to recognize a few key patterns and know what each piece of an expression is doing.

Five kinds of objects in R code

When you look at a line of R code, you'll mostly see these five things:

1. **Data tables** - start with a capital letter (e.g., `BabyNames`)
2. **Functions** - always followed by an open parenthesis (e.g., `filter()`, `mean()`)
3. **Arguments** - what's *inside* the parentheses
4. **Variables** - the columns inside a data table (e.g., `name`, `sex`)
5. **Constants** - single values like `"Arjun"` or `1998`

> **Note**
>
> The capital/lowercase convention is a **style choice** that is followed by the *Data Computing* textbook; but it is not strictly enforced by R.

Spot the parts

```
BabyNames %>%
  filter(name == "Arjun") %>%
  summarise(total = sum(count))
```

Try to identify:

- **Data table?**
- **Functions?**
- **Variables?**
- **Constant?**
- **Named argument?**

This single command counts the total number of babies named "Arjun" in the data: **5,578**.

Spot the parts

```
BabyNames %>%
  filter(name == "Arjun") %>%
  summarise(total = sum(count))
```

Try to identify:

- **Data table?** `BabyNames` (starts with capital, no parentheses after it)
- **Functions?** `filter`, `summarise`, `sum` (each followed by an open parenthesis)
- **Variables?** `name`, `count` (lowercase, inside parentheses)
- **Constant?** `"Arjun"` (in quotes)
- **Named argument?** `total = sum(count)` (single `=` inside the function call)

This single command counts the total number of babies named "Arjun" in the data: **5,578**.

Objects, revisited

An **object** is anything in R that has a name you can refer to later.

```
age_cutoff <- 21
my_numbers <- c(3, 7, 12, 4)
greeting   <- "hello, world"
```

Once an object exists, you can use its name anywhere you'd otherwise type the value:

```
age_cutoff + 5
```

```
[1] 26
```

```
mean(my_numbers)
```

```
[1] 6.5
```

Why bother naming things?

Compare:

```r
# Hard to update, hard to read:
filter(students, age >= 21 & gpa >= 3.5 & credits >= 60)
```

vs.

```r
# Easy to read and change:
age_min     <- 21
gpa_min     <- 3.5
credit_min  <- 60

filter(students,
       age     >= age_min,
       gpa     >= gpa_min,
       credits >= credit_min)
```

Good names make your code **readable** and **easy to change later**.

Example of naming conventions from *Data Computing*

Table 1: Naming conventions from *Data Computing*.

Object type	Convention	Example
Data tables	Capitalized	`BabyNames`, `Penguins`
Variables / columns	lowercase	`name`, `body_mass_g`
Constants you create	lowercase, snake_case	`age_cutoff`, `gpa_min`

> **Warning**
>
> Object names **must** start with a letter and can only contain letters, numbers, `_`, and `.`. No spaces, no special characters, no starting with a number. This is a **requirement**, not a convention.

Functions

What is a function?

A **function** is a reusable chunk of code that takes some input, does something with it, and gives you back an output.

You've already used many functions:

```
sqrt(16)
```

```
[1] 4
```

```
mean(c(1, 2, 3, 4, 5))
```

```
[1] 3
```

```
c(1, 8, 4)
```

```
[1] 1 8 4
```

Every function call has the same three key elements: a **name**, followed by **parentheses**, with **arguments** inside.

Anatomy of a function call

```
function_name(argument1, argument2, name = value, ...)
```

- The **name** tells R *which* function to use.
- The **arguments** tell the function *what to work on* and *how*.
- Arguments are separated by commas.
- Some arguments are **named** (`name = value`); others are matched by position.

```
# Positional: round(x, digits)
round(3.14159, 2)
```

```
[1] 3.14
```

```
# Named: order doesn't matter
round(digits = 2, x = 3.14159)
```

```
[1] 3.14
```

Named vs. positional arguments

```r
# Both of these work the same way:
seq(1, 10, 2)
```

```
[1] 1 3 5 7 9
```

```r
seq(from = 1, to = 10, by = 2)
```

```
[1] 1 3 5 7 9
```

> A good habit
>
> - Use **positional** arguments for the obvious ones (the data, the first input).
> - Use **named** arguments for everything else.
>
> Your future self will thank you when reading the code six weeks from now.

Getting help with a function

If you don't remember what arguments a function takes:

```r
?mean       # opens the help page
help(mean)  # same thing
??mean      # broader search if you're unsure of the name
```

Every help page has the same sections: **Description**, **Usage**, **Arguments**, **Value**, **Examples**.

Functions return values

Every function gives you back a **value**. You can store that value with `<-`:

```r
x <- sqrt(16)
x
```

```
[1] 4
```

```r
m <- mean(c(2, 4, 6, 8))
m
```

```
[1] 5
```

If you don't assign the result, R just prints it to the console and it's gone. The result will not be stored anywhere for you to access in the future.

> ⚠ Example
>
> ```r
> mean(c(2, 4, 6, 8))   # prints 5, but doesn't save it anywhere
> ```
>
> If you want to use that 5 later, you need `m <- mean(c(2, 4, 6, 8))`.

Functions inside functions

The output of one function can become the input to another:

```r
sqrt(mean(c(1, 4, 9, 16, 25)))
```

```
[1] 3.316625
```

R works **inside-out**:

1. First `c(1, 4, 9, 16, 25)` makes a vector.
2. Then `mean(...)` of that vector is `11`.
3. Then `sqrt(11)` is `3.317`.

This nesting can get unreadable fast. Later on, we'll see a cleaner way (the **pipe**).

Packages

What is a package?

Base R has a lot of features, but many useful tools live in **packages**.

A **package** is a bundle of:

- functions

- data sets
- documentation

Packages are written by someone in the R community and made available for everyone to use.

There are over **23,000** packages on CRAN (the main R package repository).

Installing vs. loading

There are **two steps** to using a package.

> ❗ Install once, load every session
>
> - **Install** with `install.packages("packagename")` - do this **once** per computer.
> - **Load** with `library(packagename)` - do this **every time** you start a new R session.

```
# Run this once, in the Console (NOT in your script):
install.packages("tidyverse")

# Run this at the top of every script or .qmd that uses it:
library(tidyverse)
```

Quotation marks matter: `install.packages("tidyverse")` (string), but `library(tidyverse)` (object).

Where do I install packages?

> 💡 Run `install.packages()` in the **Console**, not in your script.
>
> You only need to install once, so it doesn't belong in a script that runs every time.

If your script needs a package, just put `library(packagename)` at the top.

What happens when you load a package?

```
library(tidyverse)
```

```
 Attaching core tidyverse packages        tidyverse 2.0.0
dplyr     1.1.4      readr     2.1.5
forcats   1.0.0      stringr   1.5.1
ggplot2   3.5.1      tibble    3.2.1
lubridate 1.9.4      tidyr     1.3.1
purrr     1.0.4
 Conflicts                  tidyverse_conflicts()
dplyr::filter() masks stats::filter()
dplyr::lag()    masks stats::lag()
```

These messages are **fine**. They mean: "Here are the packages I just loaded, and here are the function names that now point to a new place."

The tidyverse

What is the tidyverse?

The **tidyverse** is a *collection* of R packages that share a common philosophy and work well together.

When you run `library(tidyverse)`, you load all of them at once:

- **`ggplot2`** - making plots
- **`dplyr`** - manipulating data tables
- **`tidyr`** - reshaping data
- **`readr`** - reading data files
- **`tibble`** - a nicer kind of data frame
- **`stringr`** - working with text
- **`forcats`** - working with categorical data
- **`purrr`** - applying functions across collections

We will use almost all of them this semester.

The tidyverse hex stickers

Each tidyverse package has its own hex sticker - a fun tradition in the R community.

Why use the tidyverse?

Two reasons:

1. **Consistency.** Tidyverse functions are designed to work the same way:
    - First argument is always the data.
    - Output is always a data frame (a "tibble").
    - You can chain them together cleanly (more on that later).
2. **Readability.** Tidyverse code reads almost like English.

We'll spend most of the semester learning to *think* in this style.

A taste of what's coming

Don't worry about understanding every line - just notice how **readable** this is:

```
library(palmerpenguins)

penguins |>
  filter(!is.na(body_mass_g)) |>
  group_by(species) |>
```

```r
  summarize(
    n           = n(),
    avg_mass_g  = mean(body_mass_g)
  )
```

```
# A tibble: 3 x 3
  species       n avg_mass_g
  <fct>     <int>      <dbl>
1 Adelie      151      3701.
2 Chinstrap    68      3733.
3 Gentoo      123      5076.
```

This is the **dplyr** package in action. We'll cover it in detail later this week.

ggplot2: a grammar of graphics

`ggplot2` (part of the tidyverse) is the most popular plotting package in R.

It's built on the idea of a **grammar**: you build up a plot by adding layers.

Every `ggplot2` plot has at least three pieces:

1. **Data** - a data frame.
2. **Aesthetics** (`aes(...)`) - which variables map to which parts of the plot (x, y, color, fill, etc.).
3. **Geometry** (`geom_*()`) - what *kind* of plot (points, bars, lines, histograms…).

Histograms

A **histogram** shows the *distribution* of a single numeric variable by bucketing values into bins and counting how many fall in each bin.

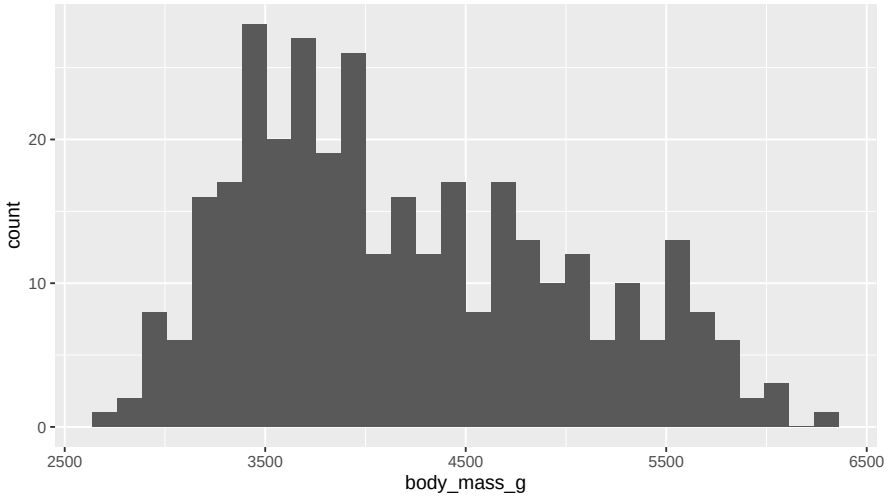
It's the right tool when you want to ask:

- What values are typical?
- Are there outliers?
- Is the distribution symmetric? Skewed? Bimodal?

In `ggplot2`, you make a histogram with `geom_histogram()`.

A basic histogram

```
ggplot(penguins, aes(x = body_mass_g)) +
  geom_histogram()
```



Reading that code

```
ggplot(penguins, aes(x = body_mass_g)) +
  geom_histogram()
```

Breaking it down:

- `ggplot(penguins, ...)`: start a plot using the `penguins` data.
- `aes(x = body_mass_g)`: map the variable `body_mass_g` to the **x-axis**.
- `+ geom_histogram()`: add a histogram layer.

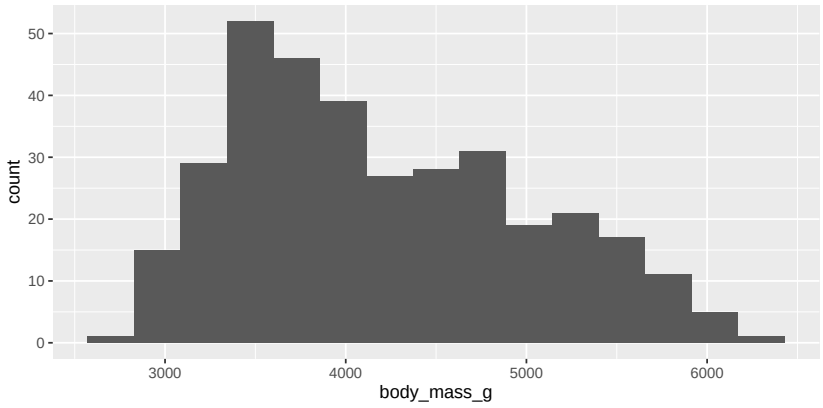
> **Note**
>
> In `ggplot2`, you add layers with `+`, not with `%>%` or `|>`.

Controlling the bins

R will warn you: *"stat_bin() using bins = 30. Pick better value with* ***binwidth****."*

You can control the number of bins two ways:

```
ggplot(penguins, aes(x = body_mass_g)) +
  geom_histogram(bins = 15)
```




```
# Or specify the width of each bin:
ggplot(penguins, aes(x = body_mass_g)) +
  geom_histogram(binwidth = 250)
```

Why the bin choice matters

⚠ Too few bins

You lose detail and the shape gets oversmoothed.

Too many bins

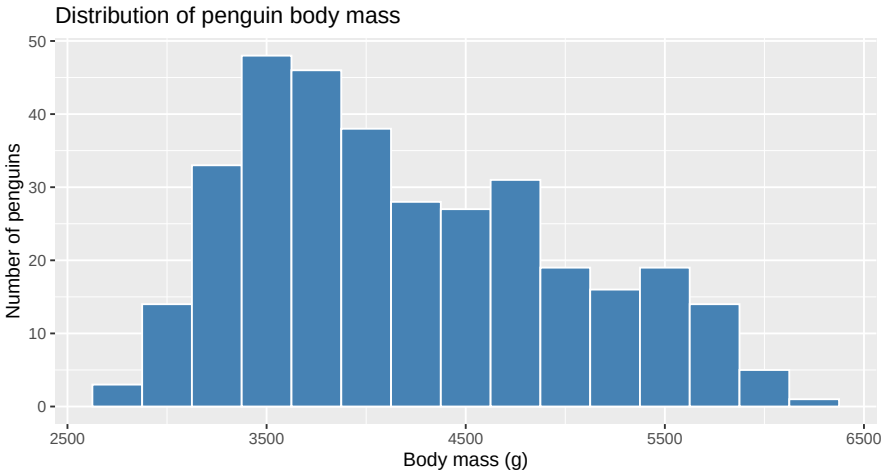
You see noise and the shape gets jagged and hard to read.

There's no single "right" answer. **Try a few**, and pick what tells the clearest story about your data.

Making it nicer

Add fill, color, and labels:

```
ggplot(penguins, aes(x = body_mass_g)) +
  geom_histogram(binwidth = 250, fill = "steelblue", color = "white") +
  labs(
    title = "Distribution of penguin body mass",
    x = "Body mass (g)",
    y = "Number of penguins"
  )
```



Anatomy of that command

Table 2: Anatomy of a basic histogram in ggplot2.

Piece	What it does
`ggplot(penguins, aes(x = body_mass_g))`	Use the penguins data; put `body_mass_g` on x
`geom_histogram(...)`	Add the histogram layer
`binwidth = 250`	Each bar is 250 g wide
`fill = "steelblue"`	Bar color
`color = "white"`	Outline color
`labs(...)`	Title and axis labels

> **Tip**
>
> You can run `?geom_histogram` in the Console to see *every* option.

Activity

Activity: your first histogram

You'll work on this individually or with a neighbor for 20 minutes. We'll regroup at the end.

Goal: Make a histogram of a real dataset and tweak it.

Setup:

1. Download the `day3-activity.R` starter file from Canvas.
2. Open RStudio.
3. Go to `File > Open File...` and open the script you downloaded.

Step 1 - Load packages

At the top of your script, type and run:

```
library(tidyverse)
library(palmerpenguins)
```

If `palmerpenguins` isn't installed, run `install.packages("palmerpenguins")` in the **Console** first.

Step 2 - Look at the data

```
# Pick ONE of these to peek at the data:
glimpse(penguins)
View(penguins)
?penguins
```

What variables are in this data set? What does each row represent?

Step 3 - Make a default histogram

Pick a numeric variable from `penguins` (e.g., `flipper_length_mm`, `bill_length_mm`, `bill_depth_mm`).

```r
ggplot(penguins, aes(x = flipper_length_mm)) +
  geom_histogram()
```

Step 4 - Tweak your histogram

Make at least **three** changes to your histogram:

- Try a different `binwidth` or `bins` value.
- Change the `fill` and `color` of the bars.
- Add a title and axis labels with `labs(...)`.

Step 5 - Comment

Open a comment block in your script and write 2–3 sentences answering:

```r
# What does the shape of your histogram tell you about the variable?
# Did changing the binwidth change the story?
```

To turn in:

Save your script (`Cmd/Ctrl + S`). Upload the `.R` file to the Canvas assignment page.

> Stuck?
>
> - Read the error message - it's usually a typo, missing `+`, or unmatched `(`.
> - Ask a neighbor.
> - Wave me down.

Activity debrief

Some things to look out for:

- **Forgot + between layers**: R thinks the line is finished.
- **Used = instead of <-**: works in some places, but breaks elsewhere.
- **Typo in a variable name**: R doesn't guess. The spelling and capitalization must match exactly.
- **Package not loaded**: `Error: could not find function "ggplot"`: run `library(tidyverse)`.

Wrap-up: what we covered today

- Recognize the five kinds of objects in R code.
- Read and call functions, including named and positional arguments.
- Install and load **packages**.
- Recognize the **tidyverse** as a coherent collection of packages.
- Build a **histogram** in `ggplot2` from scratch.

Wrap-up & looking ahead

Tomorrow: Quarto basics, writing reproducible documents.

Upcoming Deadlines

- **Wed 5/20, 11:59 p.m.** - HW 0: Setup
- **Thurs 5/21, before class** - DC §4.1–4.7
- **Fri 5/22, in class** - Quiz 1
  - Quiz 1 Practice
- **Fri 5/22, 11:59 p.m.** - HW 1: Basic R & Plots

Reach out

Stuck? Office hours, or email me (cgc5478@psu.edu) or Xinyue (xpw5228@psu.edu).

Reference

Syllabus · AI policy · Schedule · *Data Computing*